10. Perform moving of an image from one place to another.

Aim:

To perform image processing by moving an image from one position to another using the OpenCV library.

Procedure:

- Import the required libraries such as OpenCV (cv2) and NumPy.
- Read the input image using the cv2.imread() function.
- Create the required kernel/parameters for image processing.
- Apply the required image-processing operation using the appropriate OpenCV function.
- Display the original and processed images using cv2.imshow() and close the windows using cv2.destroyAllWindows().

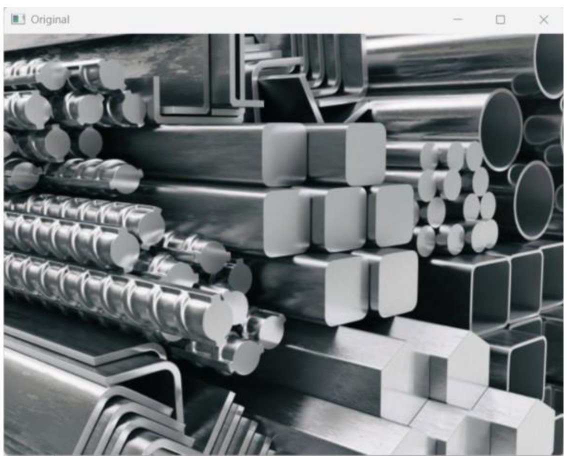
Code:

```
import cv2

import numpy as np



# Read the image from Downloads folder

image =

cv2.imread("/Users/michellejoanna/Downloads/meee.jpeg")



# Check if the image is loaded successfully

if image is None:

    print("Image not found!")

    exit()
```

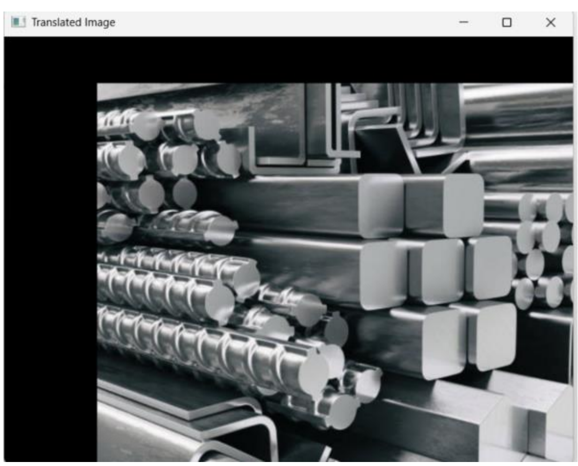
```
# Get image dimensions
rows, cols = image.shape[:2]

# Translation values
tx = 100   # Move 100 pixels to the right
ty = 50    # Move 50 pixels downward

# Translation Matrix
M = np.float32([
    [1, 0, tx],
    [0, 1, ty]
])

# Apply Translation
translated = cv2.warpAffine(image, M, (cols, rows))

# Display Images
cv2.imshow("Original Image", image)
cv2.imshow("Translated Image", translated)

# Wait until a key is pressed
```

```
cv2.waitKey(0)


# Close all windows

cv2.destroyAllWindows()
```

Sample Input



Sample Output



Result:

The input image was successfully moved from one position to another using image translation with OpenCV.